

Lecture 6: From Risk Management To Portfolio Diversification

Phase 2: Risk Management Science

OSALP 2025-04

Nurturing Future Risk Managers and Actuaries

Department of Statistics and Data Science
The Chinese University of Hong Kong
Gifted Education Fund Programme

2026

Key Definition / Result: Central Question

How do we move from vague uncertainty in real life to mathematical tools that help us make better decisions?

1. **Risk management foundations:** objectives, risk statements, risk cycle, treatment, monitoring, and reporting.
2. **Probability basics:** frequency, severity, random variables, probability mass functions, probability density functions, expectation, and standard deviation.
3. **Co-movement of risks:** joint distributions, covariance, correlation, and the interpretation of dependence.
4. **Portfolio application:** diversification, two-asset portfolios, correlation effects, short selling, and the efficient frontier.

Part A. Risk Management Foundations

Opening Problem: Planning A High-Stakes School Event

Cognitive Trigger / Opening Problem

A school plans an outdoor science day. The objectives are:

safe + **on time** + **within budget** + **educationally meaningful**

Key Definition / Result: Uncertainties

- ▶ Rainstorm or extreme heat.
- ▶ Bus delay or route closure.
- ▶ Student health issue.
- ▶ Equipment failure.

Investigation / Activity: Decision Questions

1. Which risks should we address first?
2. What actions should be taken before the event?
3. How should decisions be communicated?

Pause-and-Think

Can we completely eliminate all uncertainty while still hosting the event?

Worked Example: Fictional Exam-Style Hook**Question:** What is risk?**100 marks****Answer:** **This is risk.****Key Definition / Result**

In this lecture, **risk** means uncertainty that may affect our objectives, especially uncertainty linked to bad outcomes.

- ▶ The same uncertainty may be important or unimportant depending on the objective.
- ▶ Mathematics helps us describe uncertainty, but risk management asks what we should do about it.
- ▶ Therefore, risk is not only a probability concept. It is a decision concept.

Key Definition / Result

Risk management is a structured process for making decisions under uncertainty.

1. Protect objectives and value.

- It protects safety, learning outcomes, time, cost, and reputation.

2. Improve decisions.

- It makes trade-offs explicit instead of relying only on feelings.

3. Enable governance.

- It assigns ownership, records justifications, and monitors outcomes.

Common Mistake / Pitfall

Risk management is not “remove all risk”. It is about controlling risk to an acceptable level.

Risk Appetite Versus Risk Tolerance

Key Definition / Result: Risk Appetite

Risk appetite is the amount and type of risk an organisation is willing to take in pursuit of its objectives.

Working Example

The school may accept a small chance of schedule delay if the outdoor activity is educationally valuable.

Key Definition / Result: Risk Tolerance

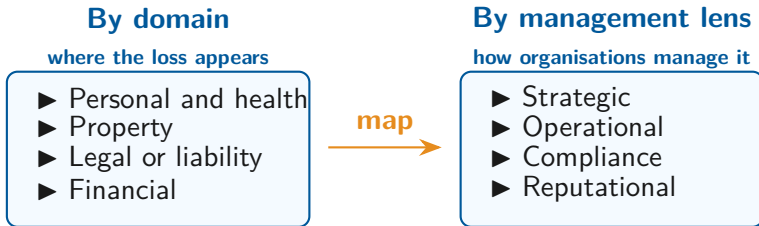
Risk tolerance is the maximum acceptable level of risk in a specific area.

Boundary / Failing Example

The school may set a strict tolerance: no activity proceeds if injury risk is judged high.

Pause-and-Think

How can we turn words such as “minor”, “high”, and “acceptable” into definitions that different people can apply consistently?



Technology Check: Interpretation

A good risk discussion often uses both lenses: first describe the real-world domain, then translate it into a management category.

Key Definition / Result

To manage risk accurately, we must distinguish between the **immediate cause** of a loss and the **conditions** that make it more likely or severe.

Working Example: Peril

A **peril** is the actual event that directly causes a loss.

Example: A heavy rainstorm that washes out the outdoor stations.

Working Example: Hazard

A **hazard** is a condition that increases the likelihood or severity of a peril.

Example: Choosing a low-lying field with poor drainage (physical hazard).

Pause-and-Think

If a student forgets to bring their water bottle, is that a peril or a hazard for heat exhaustion?

Key Definition / Result

A clear risk statement has a cause-and-effect structure:

If ⟨event⟩ **then** ⟨impact on objective⟩.

Working Example: Clear

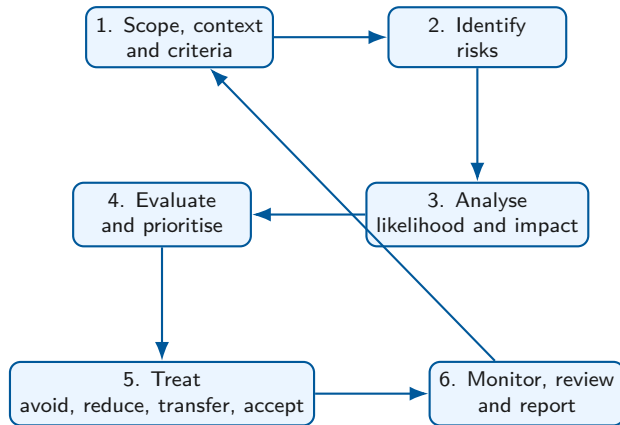
If the bus is delayed by more than 30 minutes, then the group misses Station 1 and the learning outcome is weakened.

Boundary / Failing Example: Vague

“Bad weather” is too broad. It does not say what happens, which objective is affected, or what action is needed.

Pause-and-Think

Why is “bad weather” harder to manage than “heavy rain closes the outdoor station”?



Technology Check

This cycle prevents risk management from becoming a one-time checklist.

Key Definition / Result

Before rating risks, we must define the playing field.

- ▶ **Scope:** What activity, time horizon, and stakeholders are included?
- ▶ **Context:** What internal and external factors matter?
- ▶ **Criteria:** What do Low, Medium, and High mean for likelihood and impact?

Working Example: Better Criteria

Likelihood: rare / sometimes / frequent.

Impact: minor / serious / critical.

Boundary / Failing Example: Weak Criteria

Likelihood: maybe / perhaps / probably.

Impact: bad / very bad.

Pause-and-Think

If two people interpret “perhaps” differently, can their ratings be compared fairly?

Key Definition / Result

Risk identification asks: where can the plan fail, and how would that failure affect our objectives?

- ▶ Brainstorming with prompts.
- ▶ Past-incident checklist.
- ▶ Process map or walk-through.
- ▶ Interviews with frontline users.

Investigation / Activity

In pairs, write **six** risk statements for the outdoor science event using:

If ..., then ...

Key Definition / Result

A common qualitative analysis rates each risk along two separate dimensions:

$$\text{Risk rating} \leftarrow \text{Likelihood} + \text{Impact}.$$

Key Definition / Result: Likelihood

How likely is the event to happen within the relevant time horizon?

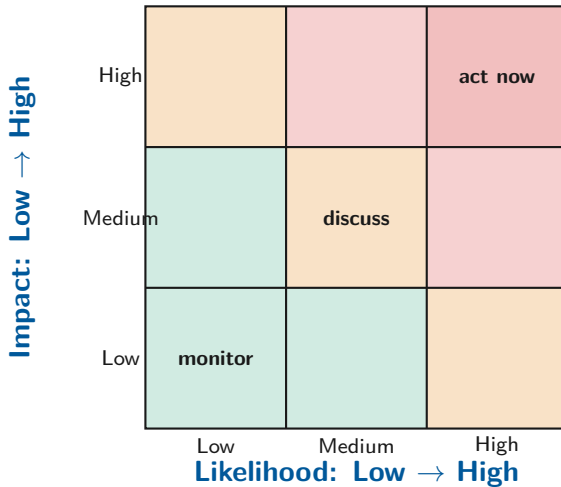
Key Definition / Result: Impact

If it happens, how serious is the effect on the objective?

Common Mistake / Pitfall

Do not rate a terrifying event as “high likelihood” merely because the consequence is severe. Likelihood and impact must be judged separately.

Step 3 Tool: Risk Heat Map



Pause-and-Think

Worked Example

Rate the following risks for the outdoor science event:

1. Heavy rain closes outdoor stations.
2. Bus delay exceeds 30 minutes.
3. Minor injury occurs during a field activity.
4. Equipment fails during a demonstration.
5. A student is missing during transition.

Pause-and-Think

How does human psychology blur the line between “how likely” and “how frightening”?

Key Definition / Result

Risk evaluation turns analysis into decisions.

Above tolerance
must treat

Near boundary
discuss trade-offs

Acceptable
monitor and record

Technology Check

A risk can be acceptable only after we know the objective, the criteria, and the treatment options. “Accept” should not mean “ignore”.

Key Definition / Result: Four Standard Responses

1. **Avoid** the activity.
2. **Reduce** likelihood or impact.
3. **Transfer/share** the risk.
4. **Accept** with justification.

Working Example: Heavy Rain Risk

If heavy rain occurs, then outdoor stations cannot run.

A sensible plan may combine reduction and partial avoidance: weather monitoring, indoor backup stations, and clear decision deadlines.

Proof / Modelling Decision Point

Choice. Choosing reduce instead of avoid

Why this choice matters. Moving everything indoors may remove rain risk, but it may also remove the educational value of an outdoor science activity.

Key Definition / Result

Inherent risk is the level of risk before controls.

Residual risk is the level of risk after controls.

- ▶ We do not approve the original risk blindly.
- ▶ We evaluate whether the remaining risk after treatment is acceptable.
- ▶ Controls must be checked, not merely written down.

Pause-and-Think

If residual risk is still above tolerance after all reasonable controls, what should a responsible decision-maker do?

Key Definition / Result

A **Key Risk Indicator** is an early warning signal that suggests a risk may be increasing.

Working Example: Useful KRIs For Transportation Risk

- ▶ Average delay in the past two weeks.
- ▶ Number of route disruption alerts.

Boundary / Failing Example: Weak KRI

“Total number of buses owned” is mostly a static fact, not an early warning signal.

Pause-and-Think

What is the difference between a lagging indicator and a leading indicator?

Key Definition / Result

A **risk register** records risk statements, ratings, treatments, ownership, deadlines, and residual risk.

Risk statement	Objective	L	I	Treatment	Owner / Due
Bus delay > 30 min ⇒ miss Station 1	Time / learning	M	M	Reduce	Team A / Fri
Heat exhaustion ⇒ medical incident	Safety	L	H	Reduce	Team B / Thu

Technology Check

A spreadsheet risk register can later become a dataset for probability modelling.

Technology Check: Lecture Bridge

Bridge. The risk register helps us describe and manage uncertainty qualitatively. But sooner or later, we want to ask quantitative questions.

How often will the event occur? If it occurs, how severe is the loss?

Key Definition / Result

This leads naturally to two quantitative modelling components:

- ▶ **Frequency:** number of events over a time horizon.
- ▶ **Severity:** loss amount given that an event occurs.

Part B. Probability As The Language Of Risk

Context	Event	Frequency	Severity
Outdoor event	Student injury	Number of injuries per event	Seriousness / medical cost
Insurance	Claim	Claims per month	Claim size
Phone warranty	Screen break	Breaks per day	Repair cost
Earthquake risk	Earthquake	Number over lifetime	Magnitude / damage

Technology Check

Probability gives us a language to model frequency and severity, instead of relying only on words such as “unlikely” or “serious”.

Key Definition / Result

A **random variable** is a quantity whose value depends on chance.

Worked Example: Coin

$X = 1$ if heads, and $X = 0$ otherwise.

Worked Example: Claim Count

N = number of claims in a month.

Worked Example: Claim Size

Y = amount paid for one claim.

Technology Check

In risk modelling, a random variable turns an uncertain real-world quantity into something we can analyse mathematically.

Key Definition / Result: Discrete Random Variable

Possible values are separate or countable.

Examples:

- ▶ Number of accidents.
- ▶ Number of claims.
- ▶ Result of a dice roll.

Key Definition / Result: Continuous Random Variable

Possible values lie on an interval.

Examples:

- ▶ Height.
- ▶ Waiting time.
- ▶ Claim size.

Proof / Modelling Decision Point

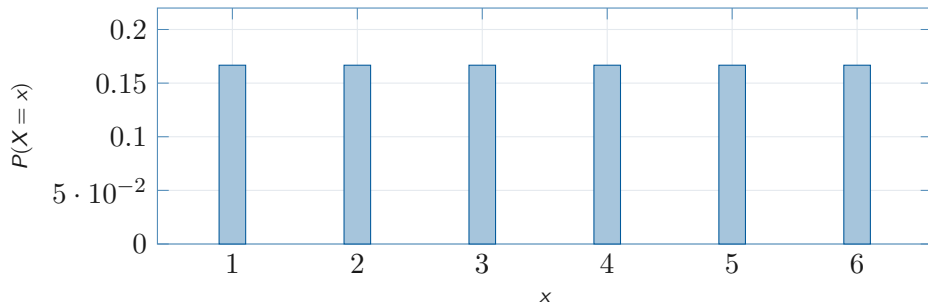
Choice. Bars versus areas

Why this choice matters. Discrete probabilities are assigned to exact values. Continuous probabilities are assigned to intervals, so we use area under a curve.

Key Definition / Result

For a discrete random variable X , the probability mass function is

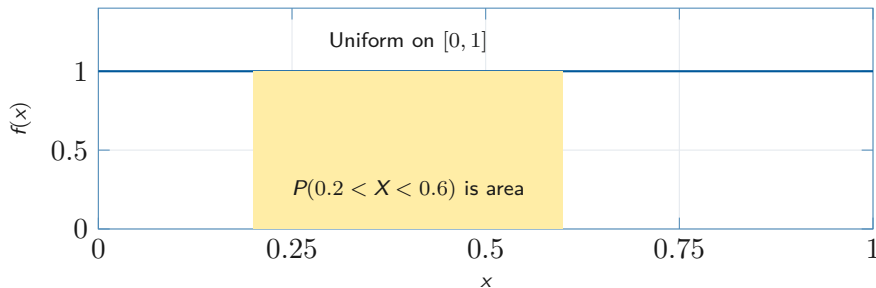
$$f(x) = P(X = x).$$



Key Definition / Result

For a continuous random variable X , the **probability density function** $f(x)$ describes density. Probabilities are areas:

$$P(a < X < b) = \int_a^b f(x) dx.$$



Working Example: Sample PMF

Data:

1, 1, 1, 1, 2, 2, 3, 3, 3, 3, 3.

Then

$$\hat{f}(1) = 4/11,$$

$$\hat{f}(2) = 2/11,$$

$$\hat{f}(3) = 5/11.$$

Boundary / Failing Example: Raw Counts Are Not Probabilities

Saying $\hat{f}(1) = 4$ is wrong because probabilities must sum to 1.

Technology Check

For continuous data, a histogram estimates the shape of the probability density function. Bin choices should be reported because they affect the appearance.

Key Definition / Result

A probability mass function or probability density function gives rich information, but it may be hard to compare whole distributions directly.

Worked Example: Expectation

Expectation is a measure of centre:

$$\mu = E(X).$$

Worked Example: Standard Deviation

Standard deviation is a measure of spread or uncertainty:

$$\sigma = \sqrt{\text{Var}(X)}.$$

Technology Check

In risk interpretation, the mean describes the typical level, while the standard deviation describes unpredictability.

Key Definition / Result

For discrete X :

$$\mu = E(X) = \sum_x xf(x).$$

For continuous X :

$$\mu = E(X) = \int xf(x) dx.$$

Technology Check

In practice, we estimate μ using the sample mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Key Definition / Result

Variance measures squared deviation from the mean:

$$\sigma^2 = E[(X - \mu)^2].$$

Standard deviation is the square root:

$$\sigma = \sqrt{\sigma^2}.$$

Worked Example

Same mean but larger standard deviation means the outcome is less predictable. In many risk contexts, this means higher uncertainty.

Worked Example

Let X be the result of a fair six-faced dice roll.

$$P(X = x) = \frac{1}{6}, \quad x = 1, 2, \dots, 6.$$

$$E(X) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \mathbf{3.5}.$$

Common Mistake / Pitfall

The expected value is not necessarily a possible single outcome. You cannot roll 3.5 on one throw.

Pause-and-Think

If 3.5 cannot occur in one trial, why is expected value still useful for pricing a repeated game?

Technology Check

Use R to compute numerical summaries quickly.

```
1 dice_outcomes <- c(1, 2, 3, 4, 5, 6)
2
3 mean(dice_outcomes)
4 var(dice_outcomes)
5 sd(dice_outcomes)
```

Common Mistake / Pitfall

R's `sd()` function returns the sample standard deviation, using $n - 1$ in the denominator.

Technology Check: Lecture Bridge

Bridge. So far, we have mostly studied one uncertain quantity at a time. But real risks often move together.

If two risks interact, we need tools for co-movement.

Worked Example

In finance, two stocks may rise and fall together. In operations, traffic delay and late arrival may be linked. In insurance, storms may cause many claims at once.

Part C. Joint Distributions, Covariance And Correlation



Key Definition / Result

A joint distribution describes two random variables together.

Worked Example: Joint PMF

For discrete X and Y :

$$f_{X,Y}(x, y) = P(X = x, Y = y).$$

Worked Example: Joint PDF

For continuous X and Y , probabilities are found over regions:

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y) \, dx \, dy.$$

Worked Example

Let $X = 1$ if the first coin is heads, and $Y = 1$ if the second coin is heads.

	$Y = 1$	$Y = 0$	Marginal X
$X = 1$	0.25	0.25	0.50
$X = 0$	0.25	0.25	0.50
Marginal Y	0.50	0.50	1.00

Investigation / Activity

Verify that X and Y are independent by comparing $P(X = x, Y = y)$ with $P(X = x)P(Y = y)$.

Worked Example

Let $X = 1$ if it is a weekday and $X = 0$ if it is a weekend.

Let $Y = 1$ if there is a traffic jam and $Y = 0$ otherwise.

	$Y = 1$ Traffic	$Y = 0$ No traffic	Marginal X
$X = 1$ Weekday	0.250	0.464	0.714
$X = 0$ Weekend	0.040	0.246	0.286
Marginal Y	0.290	0.710	1.000

Technology Check

From the table, $E[X] \approx 0.7143$ and $E[Y] = 0.2900$.

Key Definition / Result

For two random variables X and Y ,

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y].$$

The correlation coefficient is

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad -1 \leq \rho_{X,Y} \leq 1.$$

- ▶ Positive covariance: variables tend to move in the same direction.
- ▶ Negative covariance: variables tend to move in opposite directions.
- ▶ Zero covariance: no linear co-movement, but not necessarily independence.

Worked Example

Using the traffic-jam table,

$$\begin{aligned} E[XY] &= 1 \times 1 \times P(X = 1, Y = 1) \\ &= 0.2500. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Cov}(X, Y) &= E[XY] - E[X]E[Y] \\ &= 0.2500 - (0.7143)(0.2900) \\ &= \mathbf{0.0429}. \end{aligned}$$

Technology Check

The positive covariance indicates that weekdays and traffic jams are positively associated in this model.

Key Definition / Result

If X and Y have variances σ_X^2, σ_Y^2 and covariance σ_{XY} , then

$$\text{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}.$$

Technology Check

This is the mathematical engine behind diversification. The covariance term can increase or decrease total risk.

Pause-and-Think

If covariance is negative, why can combining two risky variables sometimes make the total outcome safer?

Worked Example: Example 1

PMF	$X = 0$	$X = 1$
$Y = 0$	0.25	0.25
$Y = 1$	0.25	0.25

$$\text{Cov}(X, Y) = \mathbf{0}.$$

Worked Example: Example 2

PMF	$X = 0$	$X = 1$
$Y = 0$	0.25	0.10
$Y = 1$	0.40	0.25

$$\text{Cov}(X, Y) = \mathbf{0.0225}.$$

Investigation / Activity

For each example, compute $\text{Var}(5X - 4Y)$ using the linear combination formula.

Worked Example

Suppose we observe paired data (X, Y) from a sample. R can compute means, variances, and covariance directly.

```
1 x_values <- c(rep(1, 3 + 5), rep(2, 10 + 2))
2 y_values <- c(rep(1, 3), rep(2, 5), rep(1, 10), rep(2, 2))
3
4 mean(x_values)
5 mean(y_values)
6
7 var(x_values)
8 var(y_values)
9
10 cov(x_values, y_values)
11 var(5 * x_values - 4 * y_values)
```

Technology Check

This computation prepares us for portfolio risk, where the total risk depends on variances and covariances between assets.

Technology Check: Lecture Bridge

Bridge. We now understand how to measure the co-movement of two random variables.

The next logical step is to apply this mathematical foundation to financial assets. This is where market risk and portfolio management come into play.

Part D. Portfolio Diversification

Key Definition / Result

Portfolio management asks how to allocate wealth among assets such as stocks, bonds, and derivatives.

- ▶ **Diversification:** spreading investments to reduce exposure.
- ▶ **Hedging:** using another position to offset a specific risk.
- ▶ **Optimisation:** choosing weights to balance return and risk.

Common Mistake / Pitfall

The central challenge is not only that each asset is risky. It is that assets may move together. Therefore, we must incorporate covariance and correlation.

Cognitive Trigger / Opening Problem

Suppose two investment projects are each volatile on their own.

Can a combination of the two be much safer than either one individually?

Key Definition / Result

Yes. If their returns tend to move in opposite directions, the ups and downs can partially cancel.

Pause-and-Think

If complete risk elimination is mathematically possible in a special case, why do we rarely see riskless high-yield portfolios in real markets?

Key Definition / Result

For a two-asset portfolio with weights w_1 and w_2 , where $w_1 + w_2 = 1$,

$$E[R_p] = w_1 E[R_1] + w_2 E[R_2].$$

The portfolio variance is

$$\begin{aligned}\sigma_p^2 &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 \\ &\quad + 2w_1 w_2 \text{Cov}(R_1, R_2).\end{aligned}$$

Equivalently,

$$\begin{aligned}\sigma_p^2 &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 \\ &\quad + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2.\end{aligned}$$

Stock	Average Return	Volatility
A	12%	2%
B	14%	4%

Worked Example

If a proportion p is invested in Stock A and $1 - p$ in Stock B, and the stocks are uncorrelated, then

$$\begin{aligned}\mu_{\text{port}} &= p\mu_A + (1 - p)\mu_B, \\ \sigma_{\text{port}}^2 &= p^2\sigma_A^2 + (1 - p)^2\sigma_B^2.\end{aligned}$$

Worked Example

The parameter p controls the portfolio:

- ▶ $p = 1$: full investment in Stock A.
- ▶ $p = 0$: full investment in Stock B.
- ▶ $p = 0.5$: equal investment in A and B.

If $p = 0.5$,

$$\mu_{\text{port}} = 12(0.5) + 14(0.5) = \mathbf{13\%}.$$

$$\sigma_{\text{port}} = \sqrt{0.25(2^2) + 0.25(4^2)} = \sqrt{5} = \mathbf{2.23\%}.$$

Technology Check

Even when the two stocks are uncorrelated, combining them can reduce volatility relative to holding only the more volatile stock.

Key Definition / Result

For correlated stocks:

$$\begin{aligned}\mu_{\text{port}} &= p\mu_A + (1 - p)\mu_B, \\ \sigma_{\text{port}}^2 &= p^2\sigma_A^2 + (1 - p)^2\sigma_B^2 \\ &\quad + 2p(1 - p)\sigma_{AB}.\end{aligned}$$

Worked Example

If $\sigma_{AB} = -5.0$, $p = 0.5$, $\mu_A = 10\%$, $\sigma_A = 2.5\%$, $\mu_B = 14\%$, and $\sigma_B = 4.0\%$, then

$$\mu_{\text{port}} = \mathbf{12\%}, \quad \sigma_{\text{port}} = \sqrt{3.0625} = \mathbf{1.75\%}.$$

Classical finance often presents a trade-off:

- ▶ **Stock A:** lower risk, moderate return.
- ▶ **Stock B:** higher risk, higher return.

Key Definition / Result

Portfolio thinking asks whether a mixture can produce a better risk-return balance than choosing only one asset.

Investigation / Activity: The Portfolio Idea

By holding a mixture of assets, investors may reduce unnecessary risk while keeping some expected return.

Working Example: Perfect Negative Correlation

$$\rho_{12} = -1$$

$$\sigma_p^2 = (w_1\sigma_1 - w_2\sigma_2)^2.$$

A special weight can make $\sigma_p = 0$.

Boundary / Failing Example: Perfect Positive Correlation

$$\rho_{12} = +1$$

$$\sigma_p = w_1\sigma_1 + w_2\sigma_2.$$

There is no diversification benefit.

Pause-and-Think

Why is positive correlation especially dangerous during systemic crises?

Key Definition / Result

Short selling means selling a security that one does not currently own. In portfolio weights, this allows values such as $p > 1$ or $p < 0$.

- ▶ Positive covariance is often found among firms with similar backgrounds.
- ▶ Negative covariance may arise among contrasting or competing assets.
- ▶ Short positions can sometimes be used to hedge, not merely speculate.

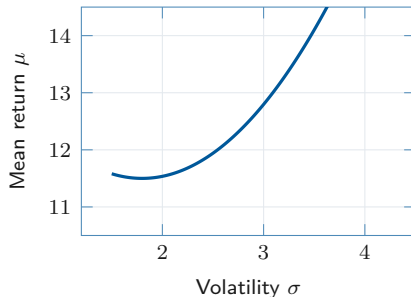
Worked Example

A portfolio with $p = 1.5$ means 150% long Stock A and 50% short Stock B.

Key Definition / Result

The efficient frontier contains portfolios that give the highest expected return for a given level of risk.

- ▶ For the same volatility, prefer higher expected return.
- ▶ For the same expected return, prefer lower volatility.
- ▶ The upper branch of the risk-return curve is efficient.



Key Definition / Result

For many assets, the Markowitz framework chooses portfolio weights to minimise risk for a target return.

- ▶ **Goal:** minimise portfolio volatility.
- ▶ **Constraint 1:** weights sum to 100%.
- ▶ **Constraint 2:** expected return equals the target return.
- ▶ **Possible extra constraint:** no short selling, so all weights are non-negative.

Technology Check

Solving the full version requires matrix algebra and computational optimisation. In this course, the main goal is to understand why covariance matters.

Investigation / Activity

Use technology to explore the following questions:

1. How does the efficient frontier change when correlation moves from $+1$ to 0 to -1 ?
2. Why can a low-return asset still be useful in a portfolio?
3. When does diversification reduce risk, and when does it fail?
4. How might transaction costs or liquidity risk reshape the efficient frontier?

Technology Check

The point is not only to calculate. The point is to connect the calculation back to risk management decisions.

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

Create a guided R investigation for gifted secondary students to explore a two-asset portfolio frontier. Use clear variable names. Let students change correlation from +1 to 0 to -1. Ask them to describe what changes visually and mathematically, and to identify when diversification works or fails.

Technology Check: Lecture Bridge

Bridge. We have covered a broad landscape, from qualitative risk registers to quantitative efficient frontiers.

Now, let us review and consolidate these concepts to ensure a holistic understanding of the entire risk management journey.

Part E. Review And Consolidation

1. Write three risk statements using “If ..., then ...” for a school event.
2. Classify each risk using:
 - the domain lens;
 - the management lens.
3. Define a three-level likelihood scale and a three-level impact scale.
4. Place six risks on a 3×3 heat map and prioritise the top two.
5. Choose a treatment option for the top two risks and justify your choices.
6. Explain inherent risk and residual risk in one sentence each.
7. Propose two KRIs for transportation risk and explain why they are early warnings.

1. Identify frequency and severity for:
 - insurance claims;
 - earthquakes;
 - phone repairs.
2. Decide whether each random variable is discrete or continuous:
 - number of accidents;
 - height;
 - claim size.
3. Given a sample of discrete data, construct the sample probability mass function.
4. Explain why probabilities must sum to 1.
5. Compute and interpret the mean and standard deviation of a simple dataset using R.
6. Explain why standard deviation can be interpreted as uncertainty or riskiness.

1. Given a joint probability mass function, find the marginal probability mass functions.
2. Compute $E[X]$, $E[Y]$, $E[XY]$, and $\text{Cov}(X, Y)$.
3. Use

$$\begin{aligned}\text{Var}(aX + bY) &= a^2\sigma_X^2 + b^2\sigma_Y^2 \\ &\quad + 2ab\sigma_{XY}\end{aligned}$$

to compute the variance of a linear combination.

4. For two assets, compute portfolio mean and volatility for $p = 0$, $p = 0.5$, and $p = 1$.
5. Explain why negative correlation can create diversification benefits.
6. Explain why positive correlation can make diversification fail during crises.

Investigation / Activity

1. If two random variables have covariance zero, are they necessarily independent? Explain.
2. Can three assets all have pairwise correlation $\rho = -0.5$? What mathematical condition must a correlation matrix satisfy?
3. Why might a portfolio with lower expected return still be preferred by a risk-averse investor?
4. How could liquidity risk, transaction costs, or model error make a mathematically efficient portfolio unattractive in practice?

Key Definition / Result**Today's big picture**

1. Risk management begins with objectives.
2. Risk statements turn vague worry into actionable language.
3. Probability quantifies frequency, severity, and uncertainty.
4. Covariance explains why risks may interact.
5. Portfolio theory uses covariance to manage market risk.

Technology Check: Moved To Tutorials / Labs

- ▶ Detailed probability density function limit definitions and integrals.
- ▶ More practice with probability mass function and probability density function calculations.
- ▶ R simulation of portfolio frontiers.
- ▶ Optional matrix formulation of Markowitz optimisation.

End Of Lecture 6